Numerical Linear Algebra
Weekly assignment #4

1. (julia only) Take a random 6-by-6 matrix A.

   ```
   n = 6
   A = rand(rng, n, n)
   ```

   We compute a Hessenberg reduction of A with

   ```
      H = copy(A)
      H, tau = LAPACK.gehrd!(H)
      Q = tril(H,-2)
      Q = Q[2:n,1:n-1]
      LAPACK.orgqr!(Q,tau)
      Q = [ 1 zeros(1,n-1); zeros(n-1,1) Q]
      H = triu(H,-1)
   ## check that
      @show opnorm( Q*H*Q'-A, 1) / opnorm( A, 1)
      @show opnorm( Q*Q' - I(n), 1)
   ```

   Let q_1 defined as

   $$q1 = \begin{pmatrix} 0.3 & 0.4 & 0.5 & -0.5 & -0.4 & -0.3 \end{pmatrix}^T.$$

   Observe that $\|q_1\|_2 = 1$.

   ```
   q1 = [ 0.3 0.4 0.5 -0.5 -0.4 -0.3]'
   @show norm(q1)
   ```

   <u>Question:</u> Reduce A to Hessenberg form such that the first column of the Q factor is q_1.

   You can first write a code for A assuming that A is Hessenberg but your final code needs to be for a general matrix A.

2. We consider A real Hessenberg. We construct $B = A^2 + 4A - 3I$. Then perform the QR factorization of B such that $Q, R = \mathrm{qr}(B)$. Finally construct $C = Q^T AQ$. Give two explanations of why C is Hessenberg. One explanation is based on using $\mathbb{C}$. The other explanation is based on looking at structure and staying in real arithmetic.

3. *Exact shift for pair of complex conjugate eigenvalue.*

   We consider A real Hessenberg with a complex conjugate eigenvalue pair $\frac{1}{4} \pm i\frac{1}{2}$. So $\frac{1}{4} \pm i\frac{1}{2}$ are two eigenvalues of A. We compute $B = A^2 - \frac{1}{2}A + \frac{5}{16}I$. Then $Q, R = \mathrm{qr}(B)$. Then $C = Q^T AQ$. Note that $\frac{1}{4} \pm i\frac{1}{2}$ are roots of the polynomial $x^2 - \frac{1}{2}x + \frac{5}{16}$.

   <u>Question:</u> What is the structure of the C? (From question 2, we know it is Hessenberg, but can we say more if we do an exact shift?)